Solving Cube Root Equations

Unit 6

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Algebra 2 • 2026

Objectives

- Solve basic cube root and rational exponent equations
- Solve multi-step cube root equations
- Apply cube root equations to real-world problems

Pro Tip

It's all the same process — isolate, undo the exponent, solve. The problems just get longer!

The Big Idea

To solve $x^{\frac{a}{b}} = c$, raise both sides to the power $\dfrac{b}{a}$.

$$\left(x^{a/b}\right)^{b/a} = c^{b/a} \quad \Longrightarrow \quad x = c^{b/a}$$

Remember

- $x^{1/3}$ is the same as $\sqrt[3]{x}$
- $x^{2/3}$ is the same as $\left(\sqrt[3]{x}\right)^2$ or $\sqrt[3]{x^2}$

The Solving Steps

Every Problem, Same Process

Step 1: Isolate the radical or power
(Get the expression with the exponent *by itself*)

Step 2: Raise both sides to the reciprocal power
(This "undoes" the radical)

Step 3: Solve for x

Watch Out!

The hardest part is Step 1! Sometimes you have to do a lot of algebra before you can cube both sides.

Solve: $\sqrt[3]{x} = 5$

Step 1: Already isolated! (Nothing else on that side)

Step 2: Cube both sides to undo the cube root

$$\left(\sqrt[3]{x}\right)^3 = 5^3$$

Step 3: Simplify

$$x = 125$$

Pro Tip

Cubing “undoes” a cube root, just like squaring undoes a square root.

Example // Rational Exponent

Solve: $x^{2/3} = 25$

Step 1: Already isolated!

Step 2: Raise both sides to the reciprocal power $\left(\frac{3}{2}\right)$

$$\left(x^{2/3}\right)^{3/2} = 25^{3/2}$$

Step 3: Simplify

$$x = \left(\sqrt{25}\right)^{3} = 5^{3} = 125$$

Pro Tip

For $25^{3/2}$: take the square root first (easier!), then cube.

Example // Isolate First

Solve: $2\sqrt[3]{x} - 5 = -4$

Step 1: Isolate the radical

$$\begin{aligned} 2\sqrt[3]{x} - 5 &= -4 \\ 2\sqrt[3]{x} &= 1 \\ \sqrt[3]{x} &= \frac{1}{2} \end{aligned}$$

Step 2: Cube both sides

$$x = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Example // Expression Inside

Solve: $(x+1)^{2/3} = 4$

Step 1: Already isolated!

Step 2: Raise both sides to the $\frac{3}{2}$ power

$$x + 1 = 4^{3/2} = \left(\sqrt{4}\right)^3 = 2^3 = 8$$

Step 3: Solve for x

$$x = 8 - 1 = 7$$

Watch Out!

Don't forget the $+1$ is still there after you undo the exponent!

Example // Multi-Step

Solve: $4\sqrt[3]{x+2}+5=-3$

Step 1: Isolate the radical (takes a few steps!)

$$\begin{aligned} 4\sqrt[3]{x+2}+5 &= -3 \\ 4\sqrt[3]{x+2} &= -8 \\ \sqrt[3]{x+2} &= -2 \end{aligned}$$

Step 2: Cube both sides

$$x+2=(-2)^3=-8$$

Step 3: Solve for x

$$x=-8-2=-10$$

Example // Radicals on Both Sides

Solve: $3\sqrt[3]{x+5} = 6\sqrt[3]{x+4}$

Step 1: Divide both sides by the smaller coefficient

$$3\sqrt[3]{x+5} = 6\sqrt[3]{x+4}$$
$$\sqrt[3]{x+5} = 2\sqrt[3]{x+4}$$

Step 2: Cube both sides

$$x+5 = 8(x+4)$$

Step 3: Solve for x

$$x+5 = 8x+32$$
$$-27 = 7x$$
$$x = -\frac{27}{7}$$

Watch Out // Cubing Coefficients

⚠ This is a Common Mistake!

When you cube both sides, don't forget to cube the coefficient too!

$$\left(2\sqrt[3]{x}\right)^3 = 2^3 \cdot x = 8x \quad \text{not } 2x$$

Example: If $\sqrt[3]{x+5} = 2\sqrt[3]{x+4}$
Cubing both sides gives:

$$x + 5 = \mathbf{8}(x+4) \quad \text{not } 2(x+4)$$

Applications // Volume of a Cube

A shipping container is a perfect cube with volume 729 cubic feet. What is the side length s?

Set up: For a cube, $V = s^3$

$$s^3 = 729$$

Solve: Take the cube root of both sides

$$s = \sqrt[3]{729} = 9 \text{ ft}$$

Pro Tip

$729 = 9^3$ is worth memorizing! It shows up a lot.

Applications // Using a Formula

Wind power is modeled by $P = 0.2v^3$. If the turbine generates 54 units of power, find the wind velocity v.

Step 1: Substitute and isolate

$$54 = 0.2v^3$$
$$270 = v^3$$

Step 2: Take the cube root

$$v = \sqrt[3]{270} \approx 6.46$$

Pro Tip

In word problems, always identify what they gave you and what you're solving for *before* you start calculating.

Special Case // No Solution?

Solve: $\sqrt{x} = -3$

Watch Out!

A **square root** can never equal a negative number!

This equation has **no solution**.

But wait... $\sqrt[3]{x} = -3$ *does* have a solution!

$$x = (-3)^3 = -27$$

Key Idea

Cube roots CAN be negative. Square roots cannot.

Key Takeaways

Remember

1. **Isolate first** — get the radical by itself
2. **Reciprocal power** — to undo $x^{a/b}$, raise to $\frac{b}{a}$
3. **Cube the coefficient too!** — $(2\sqrt[3]{x})^3 = 8x$
4. **Cube roots can be negative** — square roots can't
5. **Don't forget what's inside** — solve for x after undoing the radical

Go Sandies!

Once a Sandie, Always a Sandie